

Assignment for Week 8

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Package Loading

```
library(dplyr, warn.conflicts = F)
library(ggplot2)
library(ggmap)
library(haven)
library(httr)
library(maps)
library(mapdata)
library(rgeos)
```

```
## rgeos version: 0.3-22, (SVN revision 544)
## GEOS runtime version: 3.5.0-CAPI-1.9.0 r4084
## Linking to sp version: 1.2-4
## Polygon checking: TRUE
```

```
library(maptools)
```

```
## Loading required package: sp
## Checking rgeos availability: TRUE
```

```
library(stringr)
library(ggplot2)
library(readr)
library(ggmap)
library(maps)
library(mapdata)
library(rgeos)
library(sp)
library(rgdal)
```

```
## rgdal: version: 1.2-5, (SVN revision 648)
## Geospatial Data Abstraction Library extensions to R successfully loaded
## Loaded GDAL runtime: GDAL 2.0.1, released 2015/09/15
## Path to GDAL shared files: C:/Users/Grace/Documents/R/win-library/3.3/rgdal/gdal
## Loaded PROJ.4 runtime: Rel. 4.9.2, 08 September 2015, [PJ_VERSION: 492]
## Path to PROJ.4 shared files: C:/Users/Grace/Documents/R/win-library/3.3/rgdal/proj
## Linking to sp version: 1.2-4
```

```
library(lubridate)
```

```
##
## Attaching package: 'lubridate'
##
## The following object is masked from 'package:base':
##
##     date
```

```
library(ggrepel)
```

Assignment

- 1) 4.17 (p 249) Suppose that X and Y are random variables with the same variance. Show that $X - Y$ and $X + Y$ are uncorrelated.

Proof: We know that $\text{var}(X) = \text{var}(Y)$, and want to show $\text{cor}(X - Y, X + Y) = 0$. Since

$$\text{cor}(X - Y, X + Y) = \frac{\text{cov}(X - Y, X + Y)}{\sqrt{\text{var}(X - Y)\text{var}(X + Y)}},$$

it suffices to show $\text{cov}(X - Y, X + Y) = 0$. By the definition of covariance and the linearity of the expectation function, we have

$$\begin{aligned}\text{cov}(X - Y, X + Y) &= E[(X - Y)(X + Y)] - E[X - Y]E[X + Y] \\ &= E[X^2 - Y^2] - (E[X] - E[Y])(E[X] + E[Y]) \\ &= E[X^2] - E[Y^2] - (E[X]^2 - E[Y]^2) \\ &= (E[X^2] - E[X]^2) - (E[Y^2] - E[Y]^2) \\ &= \text{var}(X) - \text{var}(Y) \\ &= 0.\end{aligned}$$

Therefore, $X - Y$ and $X + Y$ are uncorrelated.

2) 4.18 (p 249)
 Since

$$E[W] = 0 \text{ and } \text{var}(W) = 1,$$

we have

$$E[W^2] = \text{var}(W) + E[W]^2 = 1.$$

Similarly, $E[X^2] = E[Y^2] = E[Z^2] = 1$. Since W, X, Y, Z are pairwise uncorrelated,

$$\text{cov}(W, X) = \text{cor}(W, X)\sqrt{\text{var}(W)\text{var}(X)} = 0,$$

and so

$$E[WX] = \text{cov}(W, X) + E[W]E[X] = 0.$$

Similarly, $E[WY] = E[WZ] = E[XY] = E[XZ] = E[YZ] = 0$. With this, we have

$$\begin{aligned} \text{var}(R) &= \text{var}(W + X) = E[(W + X)^2] - E[W + X]^2 \\ &= E[W^2 + 2WX + X^2] - (E[W] + E[X])^2 \\ &= E[W^2] + 2E[WX] + E[X^2] - 0^2 \\ &= 2. \end{aligned}$$

Similarly, $\text{var}(S) = \text{var}(T) = 2$.

Therefore, we have

$$\begin{aligned} \rho(R, S) &= \frac{\text{cov}(R, S)}{\sqrt{\text{var}(R)\text{var}(S)}} \\ &= \frac{1}{2} \times \text{cov}(W + X, X + Y) \\ &= \frac{1}{2} \times (E[(W + X)(X + Y)] - E[W + X]E[X + Y]) \\ &= \frac{1}{2} \times (E[WX + WY + X^2 + XY] - 0) \\ &= \frac{1}{2} \times E[X^2] \\ &= \frac{1}{2}, \end{aligned}$$

and

$$\begin{aligned} \rho(R, T) &= \frac{\text{cov}(R, T)}{\sqrt{\text{var}(R)\text{var}(T)}} \\ &= \frac{1}{2} \times \text{cov}(W + X, Y + Z) \\ &= \frac{1}{2} \times (E[(W + X)(Y + Z)] - E[W + X]E[Y + Z]) \\ &= \frac{1}{2} \times (E[WY + WZ + XY + XZ] - 0) \\ &= 0. \end{aligned}$$

Thus, the correlation coefficient of R and S is $\rho(R, S) = \frac{1}{2}$, and the correlation coefficient of R and T is $\rho(R, T) = 0$.

3) 4.19 (p 249)

First, $\text{var}(X) = E[X^2] - E[X]^2 = 1$.

Then,

$$\begin{aligned}\text{var}(Y) &= \text{var}(a + bx + cx^2) \\ &= \text{var}(a) + \text{var}(bx) + \text{var}(cx^2) \\ &= b^2 \cdot \text{var}(x) + c^2 \cdot \text{var}(x^2) \\ &= b^2 + c^2(E[X^4] - (E[X^2])^2) \\ &= b^2 + c^2(3 - 1) \\ &= b^2 + 2c^2.\end{aligned}$$

Also,

$$\begin{aligned}\text{cov}(X, Y) &= E[XY] - E[X]E[Y] \\ &= E[ax + bx^2 + cx^3] - E[X]E[a + bx + cx^2] \\ &= aE[x] + bE[x^2] + cE[x^3] - aE[X] - bE[x^2] - cE[X]E[X^2] \\ &= b.\end{aligned}$$

Therefore, the correlation coefficient $\rho(X, Y)$ is

$$\begin{aligned}\rho(X, Y) &= \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X)\text{var}(Y)}} \\ &= \frac{b}{\sqrt{b^2 + 2c^2}}.\end{aligned}$$

- 4) 4.30 (p 256) Calculate $E[X^3]$ and $E[X^4]$ for a standard normal random variable X .
 Since X is a standard normal random variable, the PDF $f_X(x)$ of X is

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}},$$

for all x in $\mathbb{R}$. Then the moment generating function $M(t)$ of X is given by the transformation

$$\begin{aligned} M(t) &= E[e^{tX}] \\ &= \int_{\forall x} \left(e^{tx} f_X(x) \right) dx \\ &= \int_{\forall x} \left(e^{tx} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \right) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(e^{-\frac{x^2}{2} + tx} \right) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(e^{-\frac{(x-t)^2}{2}} e^{\frac{t^2}{2}} \right) dx \\ &= e^{\frac{t^2}{2}} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-t)^2}{2}} dx \right) \\ &= e^{\frac{t^2}{2}}, \end{aligned}$$

the last step follows from the fact that $X - t$ satisfies the normalization requirement

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-t)^2}{2}} dx = 1.$$

Then, by $E[X^n] = \frac{d^n M(t)}{dt^n} \Big|_{t=0}$, we have

$$\begin{aligned} E[X^3] &= \frac{d^3 M(t)}{dt^3} \Big|_{t=0} \\ &= e^{\frac{t^2}{2}} (t^3 + 3t) \Big|_{t=0} \\ &= 0, \end{aligned}$$

and

$$\begin{aligned} E[X^4] &= \frac{d^4 M(t)}{dt^4} \Big|_{t=0} \\ &= e^{\frac{t^2}{2}} (t^4 + 6t^2 + 3) \Big|_{t=0} \\ &= 3. \end{aligned}$$

- 5) 4.30 (p 256) Find the third, fourth, and fifth moments of an exponential random variable with parameter λ .

Let X be an exponential random variable with the PDF

$$f_X(x) = \lambda e^{-\lambda x}$$

for all $x > 0$. Then the moment generating function $M(t)$ of X is given by the transformation

$$\begin{aligned} M(t) &= E[e^{tX}] \\ &= \int_{\forall x} (e^{tx} f_X(x)) dx \\ &= \int_0^\infty (e^{tx} \lambda e^{-\lambda x}) dx \\ &= \lambda \int_0^\infty e^{(t-\lambda)x} dx \\ &= \lambda \left. \frac{e^{(t-\lambda)x}}{t-\lambda} \right|_0^\infty \\ &= \lambda \left(0 - \frac{1}{t-\lambda} \right) \text{ for } t - \lambda < 0 \\ &= \frac{\lambda}{\lambda - t}. \end{aligned}$$

Continuously differentiating $M(t)$ gives

$$\begin{aligned} \frac{dM(t)}{dt} &= \frac{\lambda}{(\lambda - t)^2}, \\ \frac{d^2M(t)}{dt^2} &= \frac{2\lambda}{(\lambda - t)^3}, \\ \frac{d^3M(t)}{dt^3} &= \frac{6\lambda}{(\lambda - t)^4}, \\ \frac{d^4M(t)}{dt^4} &= \frac{24\lambda}{(\lambda - t)^5}, \text{ and} \\ \frac{d^5M(t)}{dt^5} &= \frac{120\lambda}{(\lambda - t)^6}. \end{aligned}$$

Therefore, the third, fourth, and fifth moments of X are

$$\begin{aligned} E[X^3] &= \left. \frac{d^3M(t)}{dt^3} \right|_{t=0} = \frac{6}{\lambda^3}, \\ E[X^4] &= \left. \frac{d^4M(t)}{dt^4} \right|_{t=0} = \frac{24}{\lambda^4}, \text{ and} \\ E[X^5] &= \left. \frac{d^5M(t)}{dt^5} \right|_{t=0} = \frac{120}{\lambda^5}. \end{aligned}$$

6) Plot the number of bombs dropped in each contemporary European country during World War I.

```
# Obtain and organize data
shape1 <-
maptools::readShapeSpatial("C:\\Users\\Grace\\neweuropemap.shp")

## Warning: use rgdal::readOGR or sf::st_read

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shape1@data$id<-rownames(shape1@data)
shape.fort<-fortify(shape1, region="id")
shape.data<-shape.fort%>%left_join(data.frame(shape1))

## Joining, by = "id"
df<-read_csv("C:\\Users\\Grace\\THOR_WW1.csv")

## Parsed with column specification:
## cols(
##   .default = col_character(),
##   id = col_integer()
## )

## See spec(...) for full column specifications.
# Plot a bar graph of number of bombs in each contemporary European countries
tableEurope <- function(x) {
  europefile <- shape.data[shape.data$NAME == x, ]
  bombsin <- sp::point.in.polygon(point.x = as.numeric(df$Lon), point.y = as.numeric(df$Lat),
                                pol.x = europefile$long, pol.y = europefile$lat)
  table(bombsin)["1"]
}

countries <- unique(unlist(shape.data$NAME))

bombsEurope1 <- mapply(countries, FUN = tableEurope)

## Warning in sp::point.in.polygon(point.x = as.numeric(df$Lon), point.y =
## as.numeric(df$Lat), : NAs introduced by coercion

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```



```

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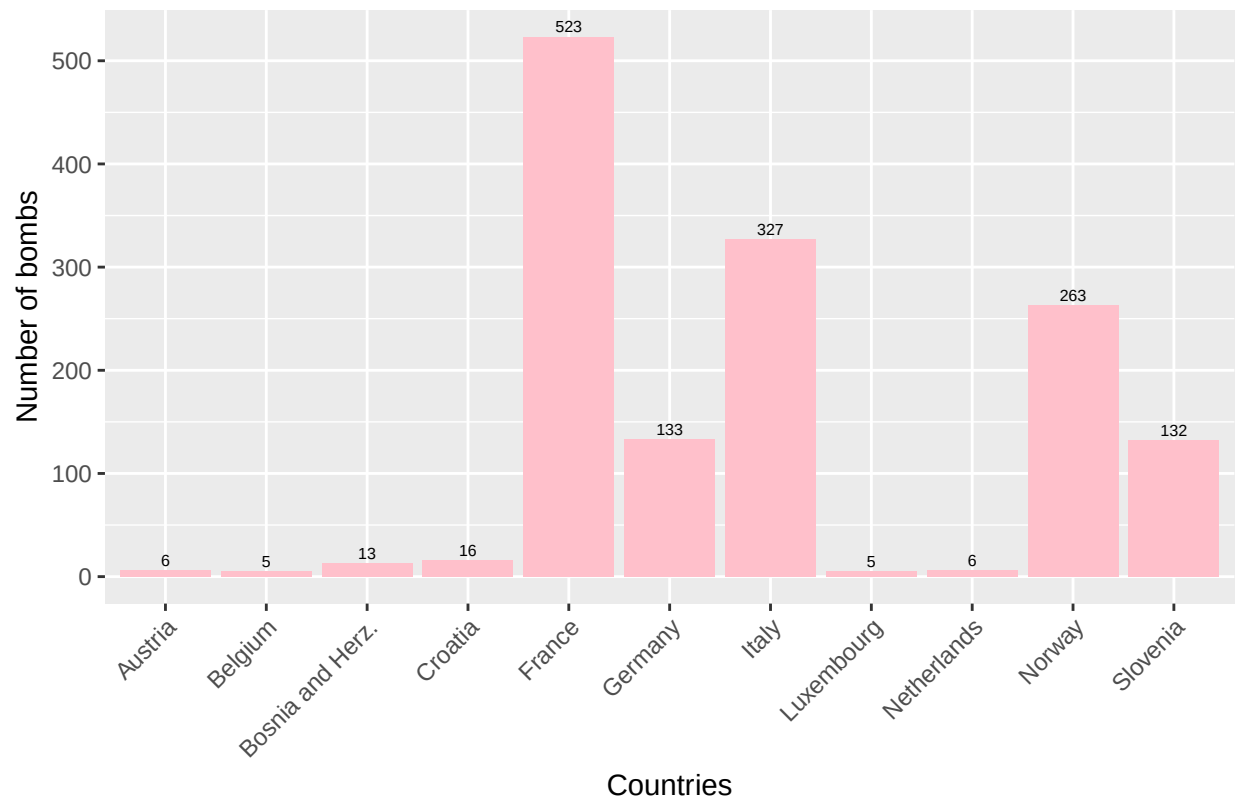
bE1 <- as.data.frame(bombsEurope1)
bE1$names <- countries
bE2 <- na.omit(bE1)

bombsplot1 <- ggplot(bE2, aes(x=names, y=bombsEurope1)) + geom_bar(stat="identity", fill="pink")

bombsplot1 + ggtitle ("Bombs in Europe during WWI")+
  theme(plot.title=element_text(hjust=0.5))+
  theme(axis.text.x = element_text(angle=45, hjust=1))+
  labs(x="Countries", y="Number of bombs")+
  geom_text(aes(label=bombsEurope1), size = 2, vjust=-0.4)

```

Bombs in Europe during WWI



- 7) Estimate the location (coordinates) of the center of gravity of bombs dropped during World War I using the Thor dataset. Each row of the Thor dataset describes many bombs being dropped. How does that average move over the course of the war? Estimate its location for 1, 3, and 6 month rolling averages.